

Multiple Choice Questions (38 Points)

I. PRELIMINARIES (1 pts)

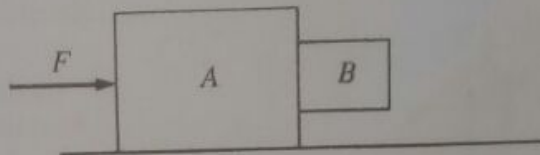
1. Two vectors are given by $\vec{A} = 4\vec{i} - 3\vec{j} + \vec{k}$ and $\vec{B} = -\vec{i} + \vec{j} + 4\vec{k}$. Find a vector $\vec{C}$ such that $\vec{A} - \vec{B} + \vec{C} = 0$

- A. $-4\vec{i} + 3\vec{j} - \vec{k}$
- B. $-5\vec{i} + 4\vec{j} + 3\vec{k}$
- C. $3\vec{i} - 2\vec{j} + 5\vec{k}$
- D. $5\vec{i} - 4\vec{j} - 3\vec{k}$
- E. $\vec{i} - \vec{j} - 4\vec{k}$

II. KINEMATICS AND DYNAMICS OF PARTICLES (5 pts)

1. For the system consisting of the two blocks shown in the figure, the minimum horizontal force F is applied so that block B does not fall under the influence of gravity. The masses of A and B are 16.0 kilograms and 4.00 kilograms, respectively. The horizontal surface is frictionless and the coefficient of friction between the two blocks is 0.50. The magnitude of F is most nearly

- A. 50 N
- B. 100 N
- C. 200 N
- D. 400 N
- E. 1,600 N



2. Just as a car starts to accelerate from its initial velocity 4 m/s with acceleration 1.4 m/s^2 , a bus moving with constant speed of 16 m/s passes it in a parallel lane. How long does it take for the car to overtake the bus?

- A. 17.1 s
- B. 8.6 s
- C. 2.9 s

D. 14.3 s

E. It will never overtake the bus

3. A cannonball is fired with a velocity v . At what angle from the ground must the cannonball be fired in order for it to hit an enemy that is at the same elevation, but a distance d away?

- A. $\sin^{-1}(v/gd)$
- B. $\sin^{-1}(gd/2v)$
- C. $\sin^{-1}(2gd/v)$

D. $(1/2)\sin^{-1}(gd/v^2)$

E. $\sin^{-1}(gd/v^2)$

4. In one-dimensional collision, a particle of mass $2m$ collides with a particle of mass m at rest. If the particles stick together after the collision, what fraction of the initial kinetic energy is lost in the collision?

- A. 0
- B. $1/4$
- C. $1/3$

D. $1/2$

E. $2/3$

5. A satellite of mass m orbits a planet of mass M in a circular orbit of radius R . The time required for one revolution is

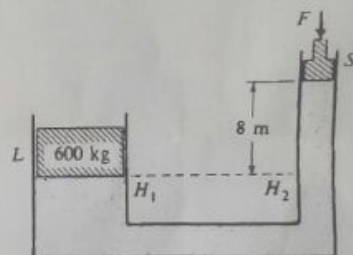
A. independent of M
 B. proportional to $\sqrt{m}$
 C. linear in R
 D. proportional to $R^{3/2}$
 E. proportional to R^2

III. FLUID MECHANICS (12 pts)

6. Find the ratio of a systolic blood pressure of 120 (in mmHg) to atmospheric pressure. Standard atmospheric pressure is $1.01 \times 10^5 \text{ Pa}$ and density of mercury is 13.6 g/cm^3

A. 1.00
 B. 1.16
 C. 0.16
 D. 0.84
 E. 6.31

7. For the hydraulic system shown in the figure, the cylinder on the left has a mass of 600 kg and cross-sectional area of 800 cm^2 . The piston on the right has cross-sectional area 25 cm^2 and negligible weight. If the apparatus is filled with oil ($\rho = 0.78 \text{ g/cm}^3$), what is the force F required to hold the mass in equilibrium?



A. 187.5 N
 B. 62.4 N
 C. 209.8 N
 D. 125.1 N
 E. 31 N

8. A diver in water picks up a lead cube of side length 10 cm. How much force is needed to lift the cube? The density of lead is approximately 11 g/cm^3 .

A. 10 N
 B. 11 N
 C. 50 N
 D. 100 N
 E. 110 N

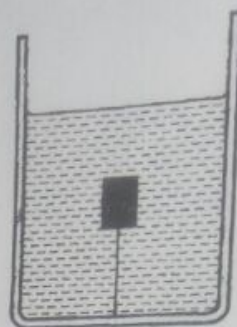
9. A boat floating in fresh water displaces 3.63 m^3 of water. What volume of water would this boat displace if it were floating in salt water of density 1024 kg/m^3

A. 3.55 m^3
 B. 3.63 m^3
 C. 3.72 m^3
 D. 7.17 m^3
 E. 0.087 m^3

10. A block of wood weighing 71.2 N and specific gravity 0.75 (or with density $3/4$ of density of water) is tied by a string to the bottom of a tank of water in order to have the block totally immersed. What is the tension in the string?

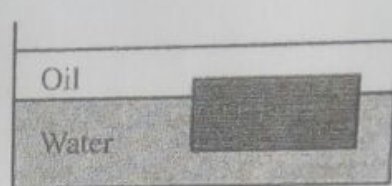
A. 23.7 N
 B. 71.2 N
 C. 94.9 N

D. 47.5 N
 E. 118.6 N



11. A layer of oil with density 800 kg/m^3 floats on top of water with density $1,000 \text{ kg/m}^3$. A block floats at the oil-water interface with $1/4$ of its volume in oil and $3/4$ of its volume in water, as shown in the figure above. What is the density of the block?

A. 200 kg/m^3
 B. 850 kg/m^3
 C. 950 kg/m^3
 D. $1,050 \text{ kg/m}^3$
 E. $1,800 \text{ kg/m}^3$



12. A spherical ball of radius R is made of a material of uniform density ρ . The ball also has a hollow spherical cavity of radius r concentric with the outer surface. If the ball just barely floats in water (density ρ_w) such that it is completely submerged except for the very top, what is the inner radius r ?

A. $R\{(\rho - \rho_w)/\rho\}^{1/3}$
 B. $R\{(\rho + \rho_w)/\rho\}^{1/3}$
 C. $R\{(\rho - \rho_w)/\rho_w\}^{1/3}$

D. $R\{\rho/\rho_w\}^{1/3}$
 E. $R\{\rho_w/\rho\}^{1/3}$

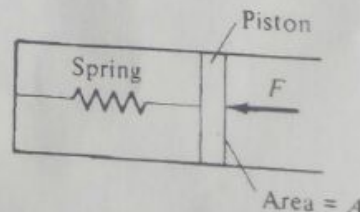
13. A vertical cylindrical tube has radius 1 cm . The tube is plugged with a stopper at the bottom end and filled with water so that the top of the water is 1 m above the bottom stopper. What is the frictional force required to hold the stopper in the tube?

A. 3.1 N
 B. 2.4 N
 C. 1.0 N

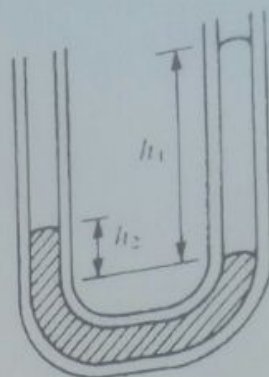
D. 0.62 N
 E. 0 N

14. A pressure gauge shown in the figure has a spring with spring constant $k = 60 \text{ N/m}$, and the area of the piston is 0.50 cm^2 . Its right end is connected to a closed container of gas at a gauge pressure of $30 \times 10^3 \text{ Pa}$. How far will the spring be compressed if the region containing the spring is in vacuum? Atmospheric pressure is $1.01 \times 10^5 \text{ Pa}$.

A. 2.5 cm
 B. 10.9 cm
 C. 8.42 cm
 D. 5.92 cm
 E. 19.3 cm



15. Assume that the two liquids in the U-shaped tube, shown in the figure, are water and oil. Compute the density of the oil if the water stands 19 cm above the interface and the oil stands 24 cm above the interface. Density of water is 1000 kg/m^3



- A. 1236 kg/m^3
- B. 5000 kg/m^3
- C. 792 kg/m^3
- D. 208 kg/m^3
- E. 1000 kg/m^3

16. An aqueduct consisting of a pipe filled completely with water passes up a hill that is 10 m high. At the bottom of the hill, a flowmeter measures the speed of the fluid to be 2.0 m/s. At the top of the hill, a flowmeter measures the speed of the fluid to be 1.0 m/s. Which is the closest to the difference in the fluid pressure between the bottom and top of the hill?

- A. $1.50 \times 10^3 \text{ Pa}$
- B. $9.85 \times 10^4 \text{ Pa}$
- C. $1.00 \times 10^5 \text{ Pa}$

- D. $1.02 \times 10^5 \text{ Pa}$
- E. $9.85 \times 10^7 \text{ Pa}$

17. An incompressible fluid of density ρ flows through a horizontal pipe of radius r and then passes through a constriction of radius $r/2$. If the fluid has pressure P_0 and velocity v_0 before the constriction, the pressure in the constriction is

- A. $P_0 - (15\rho v_0^2/2)$
- B. $P_0 - (3\rho v_0^2/2)$
- C. $P_0/4$

- D. $P_0 + (3\rho v_0^2/2)$
- E. $P_0 + (15\rho v_0^2/2)$

IV. HEAT AND THERMODYNAMICS (20 pts)

18. Why a balloon filled with hot air floats

- A. The air in balloons expands and becomes more dense than the surrounding air
- B. The air in balloons expands and becomes less dense than the surrounding air
- C. The air in balloons has no difference in density with the surrounding air
- D. None of the above

19. In thermodynamics, the internal energy of any system includes

- A. kinetic energy of translation of molecules
- B. rotation of molecules
- C. potential energy of molecules
- D. All of the above

26. Heat of any thermodynamic system is

- A. energy transferred by virtue of a temperature difference
- B. energy transferred by macroscopic work
- C. energy content of an object
- D. a temperature difference

27. During the time that latent heat is involved in a change of state

- A. the temperature does not change
- B. the substance always expands
- C. a chemical reaction takes place
- D. molecular activity remains constant

28. According to the first law of thermodynamics, applied to a gas, the increase in the internal energy during any process:

- A. equals the heat input minus the work done on the gas
- B. equals the heat input plus the work done on the gas
- C. equals the work done on the gas minus the heat input
- D. All of the above

29. The rate of heat flow by conduction through a slab does not depend upon the:

- A. temperature difference between opposite faces of the slab
- B. thermal conductivity of the slab
- C. slab thickness
- D. cross-sectional area of the slab
- E. specific heat of the slab

30. According to Ethiopian Metrological agency, today in Gondar the lowest temperature is 10°C , what is the temperature in degrees Fahrenheit and in kelvins?

- A. 20°F , 260 K
- B. 10°F , 210 K
- C. 50°F , 283 K
- D. 50°F , 200 K

31. A bimetallic strip bends as the temperature changes because

- A. the two metals have different expansion coefficients.
- B. thermal conductivity of the slab
- C. slab thickness
- D. specific heat of the slab

32. Body A has twice the mass and three times the specific heat of body B. They are supplied with equal amounts of heat. Body A experiences a temperature change ΔT . What change in temperature is experienced by body B?

A. ΔT

C. $2\Delta T/3$

B. $3\Delta T/2$

D. $6\Delta T$

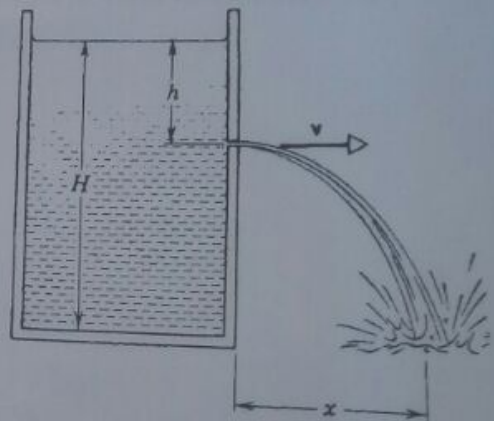
Workout Problems (12 Pts)

1. A ball of mass m is launched at 45° from the horizontal by a spring of spring constant k that is compressed by a displacement d . Show that the horizontal distance x the ball travels before landing on the ground is given by kd^2/gm .

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2. A tank is filled with water to a height H . A hole is punched in one of the walls at a depth h below the water surface, as shown in the figure. Show that the horizontal distance x from the edge of the bath where the draining water land is given by $2\sqrt{h(H-h)}$.



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3. A 0.050 kg bar of metal is heated to 200.0°C and then immersed into a beaker containing 0.400 kg of water initially at 20.0°C . If the final equilibrium temperature of the mixed system is 22.4°C , find the specific heat of the metal (Take Specific heat of water as $4186 \text{ J/kg } ^{\circ}\text{C}$)